AMBA® APB

Protocol Specification

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AMBA APB
Protocol Specification

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Release Information

The following changes have been made to this book.

Change history

Date	Issue	Confidentiality	Change
25 September 2003	A	Non-Confidential	First release, version 1.0
17 August 2004	B	Non-Confidential	Second release, version 1.0
13 April 2010	C	Non-Confidential	First release, version 2.0

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Preface

This preface introduces the *AMBA APB Protocol Specification*. It contains the following sections:

• *About this specification* on page viii
• *Feedback* on page xi

About this specification

This specification is for the *Advanced Microcontroller Bus Architecture* (AMBA) *Advanced Peripheral Bus* (APB) Protocol Specification.

Intended audience

This specification is written for hardware and software engineers who want to become familiar with the AMBA APB protocol.

Using this book

This specification is organized into the following chapters:

Chapter 1 *Introduction*

Read this for an overview of the APB protocol.

Chapter 2 *Signal Descriptions*

Read this for a description of the APB signals and their characteristics.

Chapter 3 *Transfers*

Read this for information about the typical types of APB transfer, error responses, and protection support.

Chapter 4 *Operating States*

Read this for descriptions of APB operating states and their transpositions.

Chapter 5 *Interface parity protection*

Read this for information about parity protection and check signals.

Appendix A *Signal validity*

Read this for a summary of rules when signals are valid.

Appendix B *Signal list*

Read this for signal matrix of all APB protocol signals.

Appendix C *Revisions*

Read this for a description of the technical changes between released issues of this book.

Conventions

Conventions that this specification can use are described in:

- *Typographical*
- *Timing diagrams* on page ix
- *Signals* on page ix

Typographical

The typographical conventions are:

italic Highlights important notes, introduces special terminology, denotes internal cross-references, and citations.

bold Highlights interface elements, such as menu names. Denotes signal names. Also used for terms in descriptive lists, where appropriate.

Timing diagrams

The components used in timing diagrams are explained in the figure *Key to timing diagram conventions*. Variations have clear labels, when they occur. Do not assume any timing information that is not explicit in the diagrams.

Shaded bus and signal areas are undefined, so the bus or signal can assume any value within the shaded area at that time. The actual level is unimportant and does not affect normal operation.

Clock

HIGH to LOW

Transient

HIGH/LOW to HIGH

Bus stable

Bus to high impedance

Bus change

High impedance to stable bus

Key to timing diagram conventions

Timing diagrams sometimes show single-bit signals as HIGH and LOW at the same time and they look similar to the bus change shown in *Key to timing diagram conventions*. If a timing diagram shows a single-bit signal in this way then its value does not affect the accompanying description.

Signals

The signal conventions are:

Signal level The level of an asserted signal depends on whether the signal is active-HIGH or active-LOW. Asserted means:

- HIGH for active-HIGH signals
- LOW for active-LOW signals.

Lower-case n At the start or end of a signal name denotes an active-LOW signal.

Additional reading

This section lists publications by Arm and by third parties.

See Arm Developer https://developer.arm.com/documentation for access to Arm documentation.

Arm publications

This specification contains information that is specific to this product. See the following documents for other relevant information:

- *AMBA AXI and ACE Protocol Specification* (ARM IHI 0022)
- *Arm® Realm Management Extension (RME) System Architecture Specification* (DEN 0129)

APB Revisions

APB Specification Rev E

The *APB Specification Rev E*, released in 1998, is now obsolete.

AMBA 2 APB Specification (Issue A)

The AMBA 2 APB Specification is detailed in *AMBA Specification Rev 2* (ARM IHI 0011A).

This version of the specification is referred to as APB2.

AMBA 3 APB Specification (Issue B)

The *AMBA 3 APB Protocol Specification v1.0* defines the following additional functionality:

- Wait states. See Chapter 3 *Transfers*.
- Error reporting. See *Error response* on page 3-25.

The following interface signals support this functionality:

PREADY A ready signal to indicate completion of an APB transfer.

PSLVERR An error signal to indicate the failure of a transfer.

This version of the specification is referred to as APB3.

AMBA APB Specification (Issue C)

The *AMBA APB Protocol Specification v2.0* defines the following additional functionality:

• Transaction protection. See *Protection unit support* on page 3-27.
• Sparse data transfer. See *Write strobes* on page 3-22.

The following interface signals support this functionality:

PPROT A protection signal to support both non-secure and secure transactions on APB.

PSTRB A write strobe signal to enable sparse data transfer on the write data bus.

This version of the specification is referred to as APB4.

AMBA APB Specification (Issue D)

The *AMBA APB Protocol Specification Issue D* defines the following additional functionality:
• Wakeup signaling. See *Wake-up signaling* on page 3-29.
• User signaling. See *User signaling* on page 3-30.
• Parity protection and check signals. See Chapter 5 *Interface parity protection*.

This version of the specification is referred to as APB5.

AMBA APB Specification (Issue E)

The *AMBA APB Protocol Specification Issue E* defines the following optional functionality for APB5 interfaces: • *Realm Management Extension* (RME) support. See *Realm Management Extension, RME* on page 3-28.

This version of the specification remains APB5.

Feedback

Arm welcomes feedback on this product and its documentation.

Feedback on content

If you have comments on content then send an e-mail to errata@arm.com. Give:
• The title, AMBA APB Protocol Specification
• The number, ARM IHI 0024E
• The page numbers to which your comments apply
• A concise explanation of your comments

Arm also welcomes general suggestions for additions and improvements.

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Previous issues of this document included terms that can be offensive. We have replaced these terms. If you find offensive terms in this document, please contact terms@arm.com.

Chapter 1

Introduction

This chapter provides an overview of the APB protocol. It contains the following sections: • *About the APB protocol* on page 1-14

1.1 About the APB protocol

The APB protocol is a low-cost interface, optimized for minimal power consumption and reduced interface complexity. The APB interface is not pipelined and is a simple, synchronous protocol. Every transfer takes at least two cycles to complete.

The APB interface is designed for accessing the programmable control registers of peripheral devices. APB peripherals are typically connected to the main memory system using an APB bridge. For example, a bridge from AXI to APB could be used to connect a number of APB peripherals to an AXI memory system.

APB transfers are initiated by an APB bridge. APB bridges can also be referred to as a Requester. A peripheral interface responds to requests. APB peripherals can also be referred to as a Completer. This specification will use Requester and Completer.

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Chapter 2
Signal Descriptions

This chapter describes the AMBA APB signals. It contains the following section: • *AMBA APB signals* on page 2-16

2.1 AMBA APB signals

This section describes the APB interface signals.

Some signals on the APB interface have a fixed width and some can take a variety of widths. When the width is not fixed, it is described using a property. If the property value is zero, this means the signal is not present on the interface.

Table 2-1 provides a description of the APB protocol interface signals.

Table 2-1 APB signal descriptions

Signal	Source	Width	Description
PCLK	Clock	1	Clock. **PCLK** is a clock signal. All APB signals are timed against the rising edge of **PCLK**.
PRESETn	System bus reset	1	Reset. **PRESETn** is normally connected directly to the system bus reset signal. **PRESETn** is the reset signal and is active-LOW.
PADDR	Requester	ADDR_WIDTH	Address. **PADDR** is the APB address bus. **PADDR** can be up to 32 bits wide.
PPROT	Requester	3	Protection type. **PPROT** indicates the normal, privileged, or secure protection level of the transaction and whether the transaction is a data access or an instruction access.

See *Protection unit support* on page 3-27.

PNSE Requester 1 Extension to protection type. See *Realm Management Extension, RME* on page 3-28.

PSELx Requester 1 Select.

The Requester generates a **PSELx** signal for each Completer. **PSELx** indicates that the Completer is selected and that a data transfer is required.

PENABLE Requester 1 Enable.

PENABLE indicates the second and subsequent cycles of an APB transfer.

PWRITE Requester 1 Direction.

PWRITE indicates an APB write access when HIGH and an APB read access when LOW.

PWDATA Requester DATA_WIDTH Write data.

The **PWDATA** write data bus is driven by the APB bridge Requester during write cycles when **PWRITE** is HIGH.

PWDATA can be 8 bits, 16 bits, or 32 bits wide.

Table 2-1 APB signal descriptions (continued)

Signal Source Width Description

PSTRB Requester DATA_WIDTH/8 Write strobe.

PSTRB indicates which byte lanes to update during a write transfer. There is one write strobe for each 8 bits of the write data bus. **PSTRB[n]** corresponds to **PWDATA[(8n + 7):(8n)]**.

PSTRB must not be active during a read transfer.

See *Write strobes* on page 3-22.

PREADY Completer 1 Ready.

PREADY is used to extend an APB transfer by the Completer.

PRDATA Completer DATA_WIDTH Read data.

The **PRDATA** read data bus is driven by the selected Completer during read cycles when **PWRITE** is LOW.

PRDATA can be 8 bits, 16 bits, or 32 bits wide.

PSLVERR Completer 1 Transfer error.

PSLVERR is an optional signal that can be asserted HIGH by the Completer to indicate an error condition on an APB transfer.

See *Error response* on page 3-25.

PWAKEUP Requester 1 Wake-up.

PWAKEUP indicates any activity associated

with an APB interface.

See *Wake-up signaling* on page 3-29.

PAUSER Requester USER_REQ_WIDTH User request attribute.

PAUSER is recommended to have a maximum width of 128 bits.

See *User signaling* on page 3-30.

PWUSER Requester USER_DATA_WIDTH User write data attribute.

PWUSER is recommended to have a maximum width of DATA_WIDTH/2.

See *User signaling* on page 3-30.

PRUSER Completer USER_DATA_WIDTH User read data attribute.

PRUSER is recommended to have a maximum width of DATA_WIDTH/2.

See *User signaling* on page 3-30.

PBUSER Completer USER_RESP_WIDTH User response attribute.

PBUSER is recommended to have a maximum width of 16 bits.

See *User signaling* on page 3-30.

2.1.1 Address bus

An APB interface has a single address bus, **PADDR**, for read and write transfers. **PADDR** indicates a byte address.

PADDR is permitted to be unaligned with respect to the data width, but the result is UNPREDICTABLE. For example, a Completer might use the unaligned address, aligned address, or signal an error response.

2.1.2 Data buses

The APB protocol has two independent data buses, **PRDATA** for read data and **PWDATA** for write data,. The buses can be 8 bits, 16 bits, or 32 bits wide. The read and write data buses must have the same width.

Data transfers cannot occur concurrently because the read data and write data buses do not have their own individual handshake signals.

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Chapter 3
Transfers

This chapter describes typical AMBA APB transfers, the error response, and protection unit support. It contains

the following sections:

- *Write transfers* on page 3-20
- *Write strobes* on page 3-22
- *Read transfers* on page 3-23
- *Error response* on page 3-25
- *Protection unit support* on page 3-27
- *Realm Management Extension, RME* on page 3-28
- *Wake-up signaling* on page 3-29
- *User signaling* on page 3-30

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3.1 Write transfers

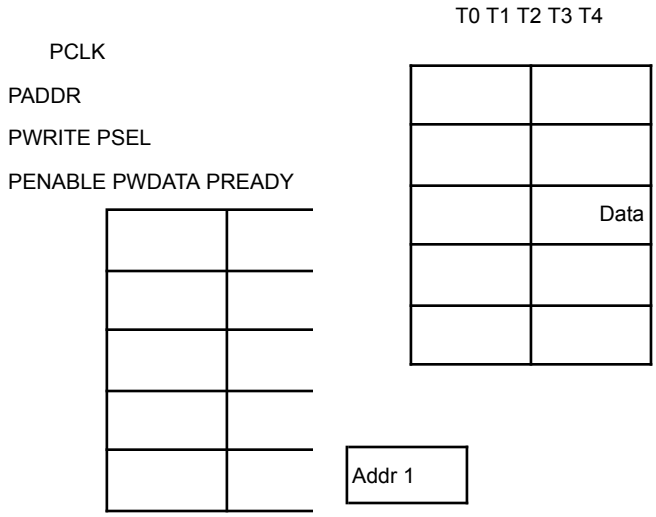
This section describes the following types of write transfer:

- With no wait states
- With wait states

All signals shown in this section are sampled at the rising edge of **PCLK**.

3.1.1 With no wait states

Figure 3-1 shows a basic write transfer with no wait states.



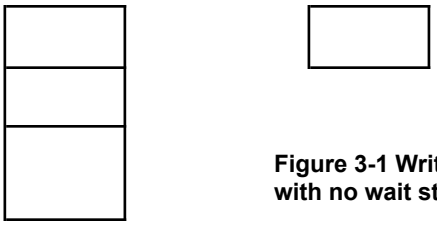


Figure 3-1 Write transfer with no wait states

The Setup phase of the write transfer occurs at T1 in Figure 3-1. The select signal, **PSEL**, is asserted, which means that **PADDR**, **PWRITE**, and **PWDATA** must be valid.

The Access phase of the write transfer is shown at T2 in Figure 3-1 where **PENABLE** is asserted. **PREADY** is asserted by the Completer at the rising edge of **PCLK** to indicate that the write data will be accepted at T3. **PADDR**, **PWDATA**, and any other control signals, must be stable until the transfer completes.

At the end of the transfer, **PENABLE** is deasserted. **PSEL** is also deasserted, unless there is another transfer to the same peripheral.

PCLK

3.1.2 With wait states

Figure 3-2 shows how the Completer can use **PREADY** to extend the transfer.

T5 T6

T0 T1 T2 T3 T4

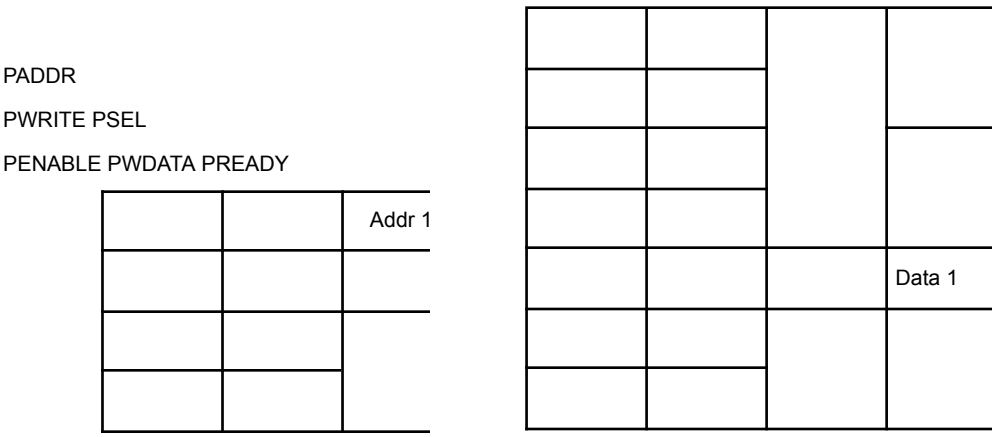


Figure 3-2 Write transfer with wait states

During an Access phase, when **PENABLE** is HIGH, the Completer extends the transfer by driving **PREADY** LOW. The following signals remain unchanged while **PREADY** remains LOW:

- Address signal, **PADDR**
- Direction signal, **PWRITE**
- Select signal, **PSELx**
- Enable signal, **PENABLE**
- Write data signal, **PWDATA**
- Write strobe signal, **PSTRB**
- Protection type signal, **PPROT**
- User request attribute, **PAUSER**
- User write data attribute, **PWUSER**

PREADY can take any value when **PENABLE** is LOW. This ensures that peripherals that have a fixed two cycle access can tie **PREADY** HIGH.

3.2 Write strobes

PSTRB enables sparse data transfer on the write data bus. Each **PSTRB** corresponds to 1 byte of the write data bus. When asserted HIGH, **PSTRB** indicates that the corresponding byte lane of the write data bus contains valid information.

There is one write strobe for each 8 bits of the write data bus, so **PSTRB[n]** corresponds to **PWDATA[(8n + 7):(8n)]**.

Figure 3-3 shows this relationship on a 32-bit data bus.

31 24 23 16 15 8 7 0

PSTRB[3] PSTRB[2] PSTRB[1] PSTRB[0]

Figure 3-3 Byte lane mapping

For read transfers, the Requester must drive all bits of **PSTRB** LOW.

3.2.1 PSTRB presence and compatibility

PSTRB is an optional signal. An APB peripheral might support a limited set of access types, which must be documented for the programmer. This means that all combinations of **PSTRB** presence might be compatible, if this document states that sparse writes are not supported.

The compatibility of **PSTRB** when connecting Requesters and Completers is described in Table 3-1.

Table 3-1 PSTRB presence and compatibility

	PSTRB Completer: signal not present	**Completer: signal present**
Requester: signal not present	Sparse writes are not supported. Compatible.	All write data byte lanes are valid for a write. Tie the **PSTRB** inputs to the **PWRITE** output from the Requester.
Requester: signal present Compatible.	Sparse writes are not supported. Compatible.	Compatible.

3.3 Read transfers

Two types of read transfer are described in this section:

- With no wait states
- With wait states

All signals shown in this section are sampled at the rising edge of **PCLK**.

3.3.1 With no wait states

Figure 3-4 shows a read transfer.

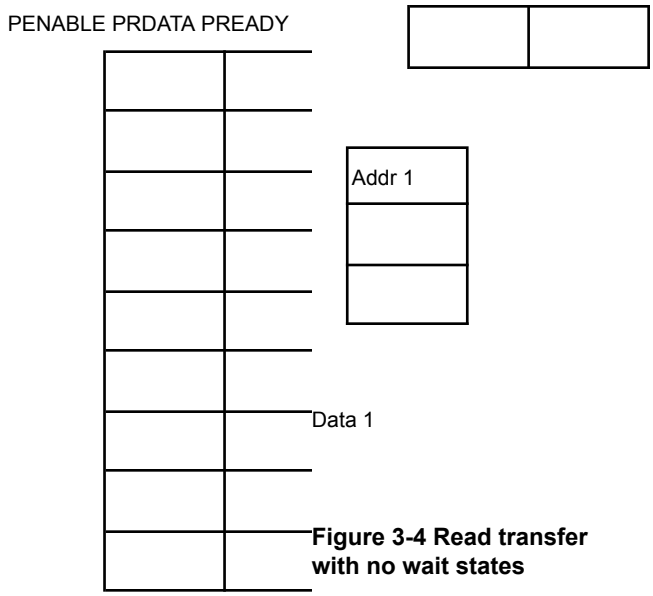


Figure 3-4 Read transfer with no wait states

The timing of the address, **PADDR**, write, **PWRITE**, select, **PSEL**, and enable, **PENABLE**, signals are the same as described in *Write transfers* on page 3-20. The Completer must provide the data before the end of the read transfer.

3.3.2 With wait states

Figure 3-5 shows how the **PREADY** signal can extend the transfer.

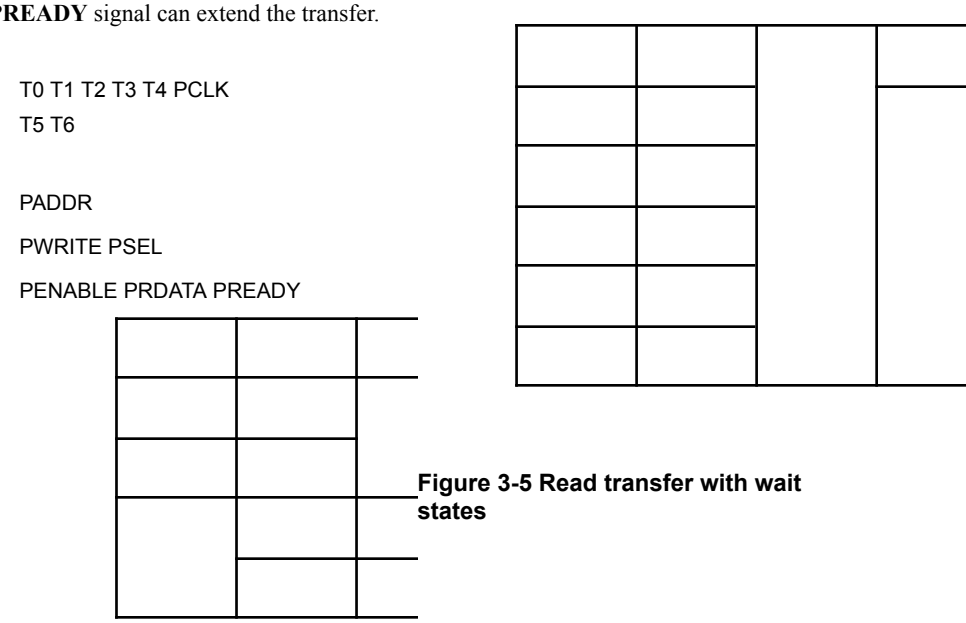


Figure 3-5 Read transfer with wait states

The transfer is extended if **PREADY** is driven LOW during an Access phase. The following signals remain unchanged while **PREADY** remains LOW:

- Address signal, **PADDR**
- Direction signal, **PWRITE**
- Select signal, **PSEL**
- Enable signal, **PENABLE**
- Protection signal, **PPROT**
- User signal, **PAUSER**

Figure 3-5 on page 3-23 shows that two cycles are added using **PREADY**. However, any number of additional cycles can be added, from zero upwards.

3.4 Error response

PSLVERR can be used to indicate an error condition on an APB transfer. Error conditions can occur on both read

and write transactions.

PSLVERR is only considered valid during the last cycle of an APB transfer, when **PSEL**, **PENABLE**, and **PREADY** are all HIGH.

It is recommended, but not required, that **PSLVERR** is driven LOW when **PSEL**, **PENABLE**, or **PREADY** are LOW.

Transactions that receive an error might or might not have changed the state of the peripheral. This is peripheral-specific and either state is acceptable.

When a write transaction receives an error, this does not mean that the register within the peripheral has not been updated.

Read transactions that receive an error can return invalid data. There is no requirement for the peripheral to drive the data bus to all 0s for a read error. A Requester which receives an error response to a read transfer might still use the data. A Completer cannot rely on the error response to prevent the reading of a value on **PRDATA**.

Completers are not required to support **PSLVERR**. Where a Completer does not include **PSLVERR**, the appropriate input to the Requester is tied LOW.

3.4.1 Write transfer

Figure 3-6 shows an example of a failing write transfer that completes with an error.

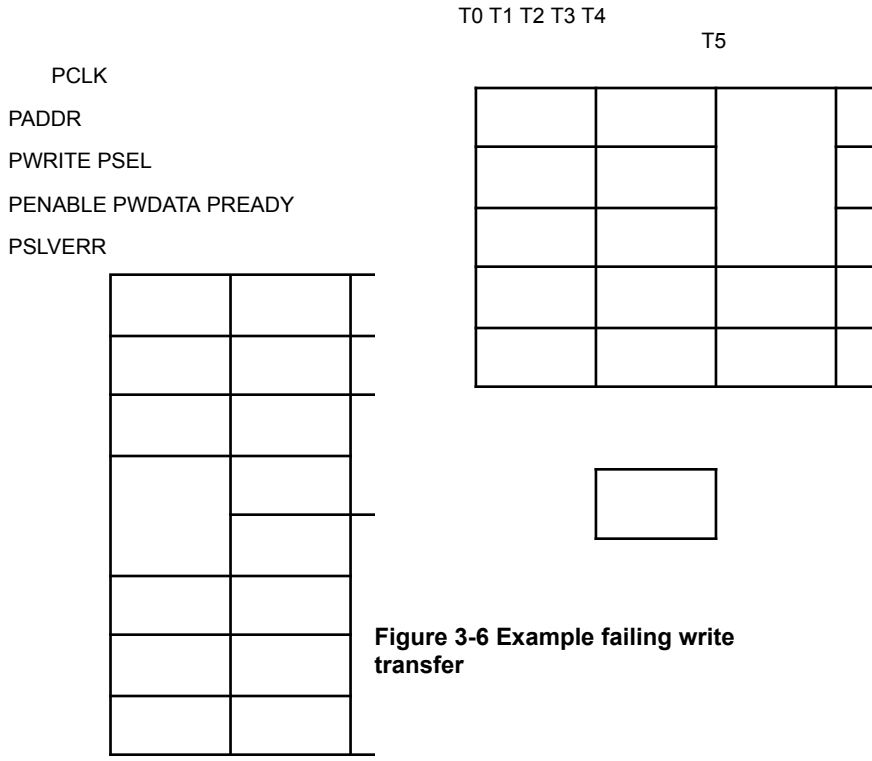


Figure 3-6 Example failing write transfer

3.4.2 Read transfer

A read transfer can also complete with an error response, indicating that there is no valid read data available. Figure 3-7 shows a read transfer completing with an error response.

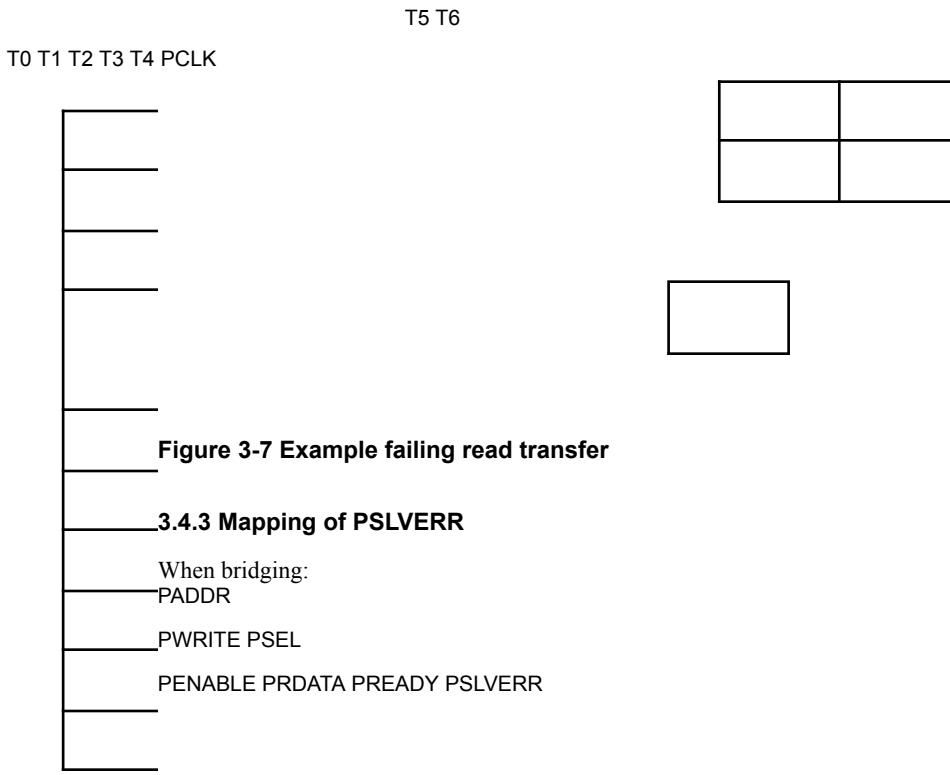


Figure 3-7 Example failing read transfer

3.4.3 Mapping of PSLVERR

When bridging:

From AXI to APB An APB error on **PSLVERR** is mapped back to **RRESP** for reads and **BRESP** for writes.

From AHB to APB An APB error on **PSLVERR** is mapped back to **HRESP** for reads and writes.

3.5 Protection unit support

To support complex system designs, it is often necessary for both the interconnect and other devices in the system to provide protection against illegal transactions. For the APB interface, this protection is provided by the **PPROT[2:0]** signals.

Table 3-2 shows the three levels of access protection with the protection level encoding.

Table 3-2 Access protection

PPROT	Protection Description	Comments
PPROT[0] Normal or Privileged	processing mode. A privileged processing mode typically has a greater level of access within a system.	• HIGH indicates privileged access.
PPROT[1] Secure or Non-secure	**PPROT[1]** is used in systems where a greater degree of differentiation between processing modes is required. • LOW indicates normal access.	• LOW indicates secure access. • HIGH indicates non-secure access.
PPROT[2] Data or Instruction	**PPROT[2]** gives an indication if the transaction is a data or instruction access. The transaction indication is provided as a hint and might not be accurate in all cases.	**Note** • LOW indicates data access. • HIGH indicates instruction access.

The primary use of **PPROT** is as an identifier for Secure or Non-secure transactions. It is acceptable to use different interpretations of the **PPROT[0]** and **PPROT[2]** identifiers.

3.5.1 PPROT presence and compatibility

PPROT is an optional signal on Requester and Completer interfaces.

Table 3-3 describes the **PPROT** compatibility when connecting a Completer to a Requester.

Table 3-3 PPROT presence and compatibility

	PPROT Completer: signal not present	**Completer: signal present**
Requester: signal not present	Compatible.	Not compatible. If fixed protection attributes are functionally correct, then the interfaces are Compatible. See Table 3-2 on page 3-27.
Requester: signal present	Compatible. The Completer has no access protection so **PPROT** can be ignored.	Compatible.

3.6 Realm Management Extension, RME

This section describes the APB protocol support for *Arm® Realm Management Extension (RME) System Architecture*.

3.6.1 Root and Realm physical address space

The RME adds Root and Realm physical address spaces into the Arm architecture. These can be used in permission checks of APB transactions.

Table 3-4 lists the additional signal added onto an APB5 interface to support RME.

Table 3-4 PNSE signal description

Signal	Width	Source	Description
PNSE	1	Requester	Extension to protection type. **PNSE** must be valid when **PSEL** is asserted.

Table 3-5 lists the permitted combinations of **PNSE** and **PPROT[1]** that determines the physical address space of the transaction.

Table 3-5 PNSE and PPROT[1] with the resulting physical address space

PNSE	PPROT[1]	Physical address space
0	0	Secure

0 1 Non-secure

1 0 Root

1 1 Realm

3.6.2 Configuration

The RME_Support property is used to indicate whether an APB interface supports RME.

True RME is supported. The **PNSE** signal is present on the interface. This can only be used on an APB5 interface.

False RME is not supported. The **PNSE** signal is not present on the interface. If the RME_Support property is not declared on the interface, it is considered False.

If **PNSE** is present on an interface, **PPROT** must also be present.

The **PNSE** signal is parity protected using the **PCTRLCHK** signal.

Table 3-6 lists the compatibility between the Requester and Completer interfaces, according to the values of the RME_Support property.

Table 3-6 RME_Support presence and compatibility

RME_Support	Completer: False	Completer: True
Requester: False	Compatible. Tie the **PNSE** input LOW.	Compatible.
Requester: True	Not compatible.	Compatible.

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3.7 Wake-up signaling

This section describes wake-up signaling used in an APB interface.

3.7.1 Introduction

The wake-up signal, **PWAKEUP**, is used to indicate any activity associated with any APB interface. **PWAKEUP** provides a glitch-free signal that can be routed to a clock controller, or similar component, to enable power and clocks to connected components.

The Wakeup_Signal property is used to indicate whether a component supports wake-up signaling:

True Wake-up signal is present.

False Wake-up signal is not present. If the Wakeup_Signal property is not declared, it is considered False.

Wake-up signaling can only be added to APB5 protocol interfaces.

3.7.2 PWAKEUP signaling

Table 3-7 describes the **PWAKEUP** signal.

Table 3-7 PWAKEUP signal description

Signal	Width	Source	Description
PWAKEUP	1	Requester	**PWAKEUP** indicates any activity associated with a Requester interface.

The rules and recommendations for **PWAKEUP** are:

- **PWAKEUP** is synchronous to **PCLK** and must be suitable for sampling asynchronously in a different clock domain. This requires **PWAKEUP** to be glitch-free. This can be achieved, for example, by being generated directly from a register, or from a glitch-free OR tree.
- **PWAKEUP** is allowed to be asserted before, during, or after the assertion of **PSELx**.
- A Completer is permitted to wait for **PWAKEUP** to be asserted, before asserting **PREADY**. The interface could deadlock if **PWAKEUP** is present but never asserted.
- **PWAKEUP** must remain asserted until **PREADY** is asserted if **PWAKEUP** and **PSELx** are HIGH in the same cycle.
- It is recommended that **PWAKEUP** be asserted at least one cycle before the assertion of **PSELx** to prevent the acceptance of a new transaction being delayed.
- It is recommended **PWAKEUP** be deasserted when no further transfers are required.
- It is permitted, but not recommended, asserting **PWAKEUP** then deasserting it without a transfer occurring.
- It is recommended that the Requester and Completer sides of a connection are clock gated together.

  If a Completer interface clock is gated independently of the Requester clock and **PWAKEUP** is used to enable the Completer clock, there is a possibility that the Setup phase of a transfer is missed by the Completer.

3.8 User signaling

This section describes user signaling in an APB interface.

3.8.1 Introduction

The users of APB protocols can encounter an application that requires the addition of signaling that is not specified in the APB protocol. User signaling defines a standard method of adding this signaling to a transaction, without defining the signal usage.

Generally, it is recommended that User signals are not used. The APB protocol interface does not define the function of these signals, causing interoperability problems if two components use the same User signals in an incompatible way.

User signaling can only be added to APB5 protocol interfaces.

3.8.2 Signaling

All signals are optional. If the associated width property is zero, the signal is not present.

Table 3-8 on page 3-30 provides a description of User signals.

Table 3-8 User signal descriptions

Signal	Width	Source	Description
PAUSER	USER_REQ_WIDTH	Requester	User-defined request attribute.

- **PAUSER** must be valid when **PSELx** is asserted.
- **PAUSER** must have the same value in the Setup and Access phase of a transfer.
- **PAUSER** must have the same value in every cycle during the Access phase of a transfer.

PWUSER USER_DATA_WIDTH Requester User-defined write data attribute.

- **PWUSER** must be valid when **PSEL** and **PWRITE** are asserted.
- **PWUSER** must have the same value in the Setup and Access phases of a transfer.
- **PWUSER** must have the same value in every cycle during the Access phase of a transfer.

PRUSER USER_DATA_WIDTH Completer User-defined read data attribute.

- **PRUSER** must be valid when **PSEL**, **PENABLE**, and **PREADY** are asserted, and **PWRITE** is deasserted.

PBUSER USER_RESP_WIDTH Completer User-defined response attribute.

- **PBUSER** must be valid when **PSEL**, **PENABLE**, and **PREADY** are asserted.

3.8.3 User signal recommendations

Where User signals are implemented, this specification does not require support for all User signals. The width of the User-defined signals is IMPLEMENTATION DEFINED and can be different for request, data, and responses.

It is recommended including provision for all User signals on a domain crossing bridge or interconnect. However, there is no requirement to include them on Completers.

It is recommended USER_DATA_WIDTH is an integer multiple of the width of the data buses in bytes to assist with data width and protocol conversion.

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Chapter 4

Operating States

This chapter describes the AMBA APB operating states. It contains the following section: • *Operating states* on page 4-34

4.1 Operating states

Figure 4-1 shows the operating states of the APB interface.

PREADY = 1
and no transfer
No transfer
IDLE
PSELx = 0

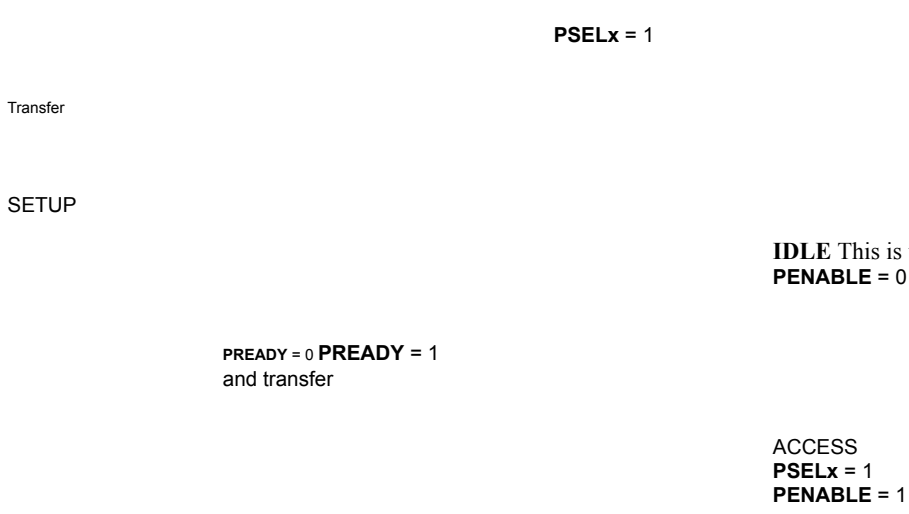

Figure 4-1 State diagram

The state machine operates through the following states:

SETUP When a transfer is required, the interface moves into the SETUP state, where the appropriate select signal, **PSELx**, is asserted. The interface only remains in the SETUP state for one clock cycle and always moves to the ACCESS state on the next rising edge of the clock.

ACCESS The enable signal, **PENABLE**, is asserted in the ACCESS state. The following signals must not change in the transition between SETUP and ACCESS and between cycles in the ACCESS state:

- **PADDR**
- **PPROT**
- **PWRITE**
- **PWDATA**, only for write transactions
- **PSTRB**
- **PAUSER**
- **PWUSER**

Exit from the ACCESS state is controlled by the **PREADY** signal from the Completer: • If **PREADY** is held LOW by the Completer, then the interface remains in the ACCESS state. • If **PREADY** is driven HIGH by the Completer, then the ACCESS state is exited and the bus returns to the IDLE state if no more transfers are required. Alternatively, the bus moves directly to the SETUP state if another transfer follows.

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Chapter 5
Interface parity protection

This chapter describes a parity scheme for detecting single-bit errors on the interface between components. It contains the following sections:

- *Protection using parity* on page 5-36
- *Configuration of interface protection* on page 5-37
- *Parity check* on page 5-38
- *Error detection behavior* on page 5-39
- *Parity check signals* on page 5-40

5.1 Protection using parity

For safety-critical applications, it is necessary to detect and correct transient and functional errors on individual wires within an SoC.

An error in a system component can propagate and cause numerous errors across connected components. *Error Detection and Correction* (EDC) is required to operate end-to-end, covering all logic and wires from source to destination.

One way to implement end-to-end protection is to employ customized EDC schemes in components and implement a simple error detection scheme between components. Between these components, there is no logic and single-bit errors do not propagate to multi-bit errors. This section describes a parity scheme for detecting single-bit errors on the interface between components. Multi-bit errors can be detected if they occur in different parity signal groups.

Figure 5-1 shows the locations where parity can be used.

Short distance wiring Long distance wiring

and routing logic

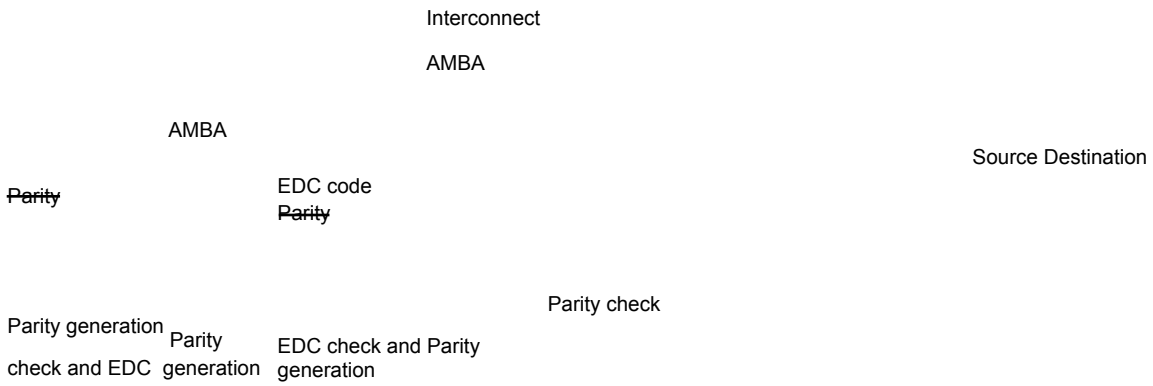

Figure 5-1 Example use of parity protection

5.2 Configuration of interface protection

The EDC scheme of the interface is defined by the Check_Type property. The following Check_Type values are defined:

False

There are no checking signals on the interface.

Odd_Parity_Byte_All

Odd parity checking is included for all signals. Each bit of the parity signal covers up to 8 bits.

If Check_Type is not declared, it is considered to be False.

Check signaling can be added to APB5 interfaces only.

5.3 Parity check

The following attributes are common to the check signals added for byte parity interface protection:

- Odd parity is used. Odd parity means that there is always an odd number of bits asserted across the interface signal and check signal. Check signals are associated with each interface signal.
- Each parity check bit covers no more than 8 bits of payload. This limitation assumes that there is a maximum of three logic levels available in the timing allowance for generating each parity bit.
- Parity signals that cover critical control signals are defined with a single parity bit. The single odd parity bit is the inversion of the original critical control signal. Critical control signals are likely to have a smaller timing allowance available.
- For a check signal that is wider than 1 bit:

— Check bit [n] corresponds to [(8n+7):8n] in the payload.

— If the payload is not an integer number of bytes, the most significant bit of the check signal covers fewer than 8-bits in the most significant portion of the payload.

- Check signals must be driven correctly in every cycle that the Check Enable term is True. See Table 5-1 on page 5-40.
- Parity signals must be driven appropriately to all the bits in the associated payload, whether or not the bits are actively used in the transfer. For example, all bits of **PWDATACHK** must be driven correctly, even if some bytes are not valid data bytes.
- If some of the signals covered by a check signal are not present on an interface, then the missing signals are assumed to be LOW.
- If none of the signals covered by a check signal are present on an interface, then the check signal is omitted from the interface.

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5.4 Error detection behavior

This specification is not prescriptive regarding component or system behavior when a parity error is detected. Depending on the system and affected signals, a flipped bit can have a wide range of effects. It might be harmless, cause performance issues, cause data corruption, cause security violations, or deadlock.

When an error is detected, the Completer can:

- Terminate or propagate the transfer.
- Correct the parity check signal or propagate the error.
- Update its memory or leave untouched.
- Signal an error response through other means, for example with an interrupt.

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5.5 Parity check signals

Check signals are synchronous to **PCLK** and must be driven correctly every cycle in which the Check Enable term is True.

Table 5-1 shows the parity check signals.

Table 5-1 Check signal descriptions

Check signal	Signals covered	Width	Granularity	Check Enable

1 1-5 **PSEL**

PADDRCHK PADDR ceil(ADDR_WIDTH/8)[a] 1-8 **PSEL**

PCTRLCHK PPROT, PWRITE, PNSE

PSELxCHK[b] **PSELx** 1 1 **PRESETn**

PENABLECHK PENABLE 1 1 **PSEL PWDATACHK PWDATA** DATA_WIDTH/8 8 **PSEL** &

PWRITE PSTRBCHK PSTRB 1 1-4 **PSEL** & **PWRITE PREADYCHK PREADY** 1 1 **PSEL** &

PENABLE

PRDATACHK PRDATA DATA_WIDTH/8 8 **PSEL** & **PENABLE** & **PREADY** & **!PWRITE**

PSLVERRCHK PSLVERR 1 1 **PSEL** & **PENABLE** & **PREADY**

PWAKEUPCHK PWAKEUP 1 1 **PRESETn PAUSERCHK PAUSER** ceil(USER_REQ_WIDTH/8) 1-8

PSEL PWUSERCHK PWUSER ceil(USER_DATA_WIDTH/8) 1-8 **PSEL** & **PWRITE**

PRUSERCHK PRUSER ceil(USER_DATA_WIDTH/8) 1-8 **PSEL** & **PENABLE** & **PREADY** & **!PWRITE**

PBUSERCHK PBUSER ceil(USER_RESP_WIDTH/8) 1-8 **PSEL** & **PENABLE** & **PREADY**

a. The function ceil() returns the lowest integer value that is equal to or greater than the input to the function. b. There is a separate check signal per **PSEL** signal as only one **PSEL** bit is routed to each Completer.

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Appendix A
Signal validity

This appendix summarizes the rules describing when signals must be valid.

A.1 Validity rules

The following signals must always be valid:

- **PSEL**
- **PWAKEUP**

The following signals must be valid when **PSEL** is asserted:

- **PADDR**
- **PPROT**
- **PNSE**
- **PENABLE**
- **PWRITE**
- **PAUSER**
- **PSTRB**
- **PWDATA**, active write data lanes only
- **PWUSER**, write only

The following signal must be valid when **PSEL** and **PENABLE** are asserted:

- **PREADY**

The following signals must be valid when **PSEL**, **PENABLE**, and **PREADY** are asserted: • **PRDATA**, read only

- **PSLVERR**
- **PRUSER**, read only
- **PBUSER**

It is recommended that signals which are not required to be valid are driven to zero.

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Appendix B
Signal list

This appendix provides a summary of all signals on the APB interface.

B.1 APB signals

Table B-1 describes the list of the APB signals. Table B-2 describes the list of APB check signals. Optional signals have a default that should be used for any un-driven inputs.

The following codes used in Table B-1 and Table B-2 are:

Y Mandatory

N Must not be present

O Optional for inputs and outputs

OO Optional for output ports, mandatory for inputs

C Conditional, must be present if the property is True

OC Optional conditional, optional but can only be present if the property is True

Table B-1 APB signals

Signal Width Default Property APB5 APB4 APB3 APB2 PCLK 1 - - Y Y YY **PRESETn** 1 - - Y Y YY **PADDR** ADDR_WIDTH - - Y Y Y Y **PPROT** 3 `0b000` - O O NN **PNSE** 1 `0b0` RME_Support C N N N **PSELx** 1 - - Y Y YY **PENABLE** 1 - - Y Y YY **PWRITE** 1 - - Y Y YY **PWDATA** DATA_WIDTH - - Y Y Y Y **PSTRB** DATA_WIDTH/8 - - O O N N **PREADY** 1 `0b1` - OO OO OO N **PRDATA** DATA_WIDTH - - Y Y Y Y **PSLVERR** 1 `0b0` - OO OO OO N **PWAKEUP** 1 - Wakeup_Signal C N N N **PAUSER** USER_REQ_WIDTH - USER_REQ_WIDTH OC N N N **PWUSER** USER_DATA_WIDTH - USER_DATA_WIDTH OC N N N **PRUSER** USER_DATA_WIDTH - USER_DATA_WIDTH OC N N N **PBUSER** USER_RESP_WIDTH - USER_RESP_WIDTH OC N N N

Table B-2 Check signal

Signal Width Property APB5 APB4 APB3 APB2 PADDRCHK ADDR_WIDTH/8 Check_Type C N N N **PCTRLCHK** 1

Check_Type C N N N **PSELxCHK** 1 Check_Type C N N N **PENABLECHK** 1 Check_Type C N N N

Table B-2 Check signal (continued)

Signal Width Property APB5 APB4 APB3 APB2 PWDATACHK DATA_WIDTH/8 Check_Type C N N N **PSTRBCHK** 1

Check_Type C N N N **PREADYCHK** 1 Check_Type C N N N **PRDATACHK** DATA_WIDTH/8 Check_Type C N N N

PSLVERRCHK 1 Check_Type OC N N N **PWAKEUPCHK** 1 Check_Type & Wakeup_Signal C N N N **PAUSERCHK**

ceil(USER_REQ_WIDTH/8) Check_Type & USER_REQ_WIDTH OC N N N

PWUSERCHK ceil(USER_DATA_WIDTH/8) Check_Type & USER_DATA_WIDTH USER_DATA_WIDTH OC N N N OC N N N

PRUSERCHK ceil(USER_DATA_WIDTH/8) Check_Type &

PBUSERCHK ceil(USER_RESP_WIDTH/8) Check_Type & USER_RESP_WIDTH OC N N N

Table B-3 shows the interface properties.

If an entry is not Y, the property must be False or undeclared for that interface type.

Table B-3 Interface properties

Property Issue introduced APB5 APB4 APB3 APB2

Check_Type D Y - - -

USER_DATA_WIDTH D Y - - -

USER_REQ_WIDTH D Y - - -

USER_RESP_WIDTH D Y - - -

Wakeup_Signal D Y - - -

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Appendix C

Revisions

This appendix describes the technical changes between released issues of this book.

Table C-1 Issue A

Change	Location
First release	-

Table C-2 Differences between issue A and issue B

Change	Location
APB signal **PREADY** added	• Table 2-1 on page 2-16 • *Write transfers* on page 3-20 • *Read transfers* on page 3-23 • *Error response* on page 3-25 • Chapter 4 *Operating States*
APB signal **PSLVERR** added	• Table 2-1 on page 2-16 • *Error response* on page 3-25

Table C-3 Differences between issue B and issue C

Change	Location
Section added listing the changes made to this specification at each revision of the document.	Appendix C *Revisions*
APB signal **PPROT** added	• Table 2-1 on page 2-16 • *Protection unit support* on page 3-27
APB signal **PSTRB** added	• Table 2-1 on page 2-16 • *Write strobes* on page 3-22

Table C-4 Changes from Issue C to Issue D

Change	Location
Added signal width properties	*AMBA APB signals* on page 2-16
Added wake-up signaling	*Wake-up signaling* on page 3-29
Added user signaling	*User signaling* on page 3-30
Added interface parity	Chapter 5 *Interface parity protection*
Included regularized terminology using Completer and Requester	Throughout the specification
Added signal validity rules	Appendix A *Signal validity*
Added signal matrix	Appendix B *Signal list*

Table C-5 Changes from Issue D to Issue E

Change	Location
Support for RME	*Realm Management Extension, RME* on page 3-28

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